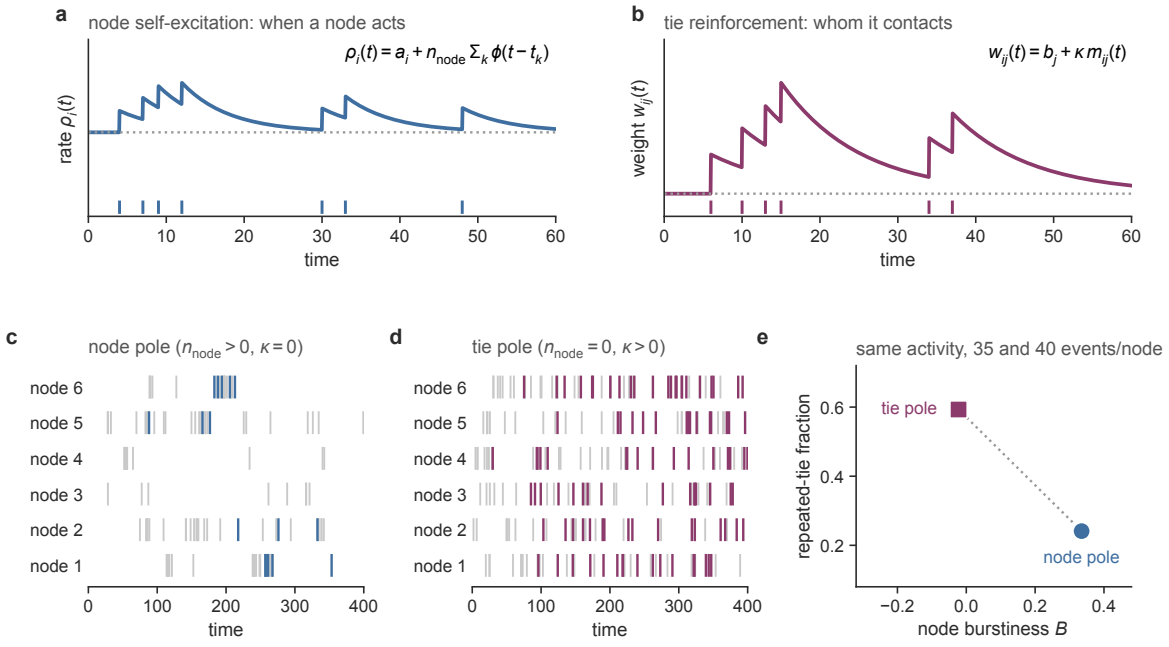
a
node self-excitation: when a node acts
$\rho_i(t) = a_i + n_{\text{node}} \Sigma_k \phi(t - t_k)$
rate $\rho_i(t)$
0
10
20
30
40
50
60
time
b
tie reinforcement: whom it contacts
$w_{ij}(t) = b_j + \kappa m_{ij}(t)$
weight $w_{ij}(t)$
0
10
20
30
40
50
60
time
c
node pole ($n_{\text{node}} > 0$, $\kappa = 0$)
node 6
node 5
node 4
node 3
node 2
node 1
0
100
200
300
400
time
d
tie pole ($n_{\text{node}} = 0$, $\kappa > 0$)
node 6
node 5
node 4
node 3
node 2
node 1
0
100
200
300
400
time
e
same activity, 35 and 40 events/node
tie pole
node pole
repeated-tie fraction
0.6
0.4
0.2
−0.2
0.0
0.2
0.4
node burstiness B